O and Gresnigt's Octonion Subalgebras Are Not Related by Any Automorphism of S: A Complete Resolution

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Abstract. A companion paper establishes that the coordinate-aligned octonion subalgebra $\mathbb{O} \subset \mathbb{S}$ and Gresnigt's octonion subalgebras $\mathbb{O}_1, \mathbb{O}_2 \subset \mathbb{S}$ exhibit identical algebraic structure — non-closure of the complement, the Cayley bimodule identity, faithfulness, irreducibility — and leaves open whether this reflects a genuine symmetry of $\mathbb{S}$ relating them, or independent recurrence of the same pattern. We resolve this completely: $\mathbb{O}$ is not related to any of Gresnigt's $\mathbb{O}_1, \mathbb{O}_2$, or $\mathbb{O}_3$ by any element of $\mathrm{Aut}(\mathbb{S}) = G_2 \times S_3$, continuous or discrete — and, by the identical argument, neither is $\mathbb{O} \cdot e_8$ related to any of their three complements. The argument requires constructing an explicit, verified basis for the 14-dimensional Lie algebra $\mathfrak{g}_2$, diagnosing and correcting a subtle structural trap along the way, and computing the full dimension of $\mathrm{der}(\mathbb{S})$ directly from the Leibniz identity, independent of any external citation, to confirm the constructed generators are complete rather than partial. The result is four complete, self-contained instances of the entire Operator Mismatch structure, not one instance viewed four ways.

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1. Background

A companion paper (*The Sedenion Split $\mathbb{S} = \mathbb{O} \oplus \mathbb{O} \cdot e_8$*) shows that non-closure, the Cayley bimodule identity, faithfulness, and irreducibility all hold identically for the complements of three independently constructed octonion subalgebras of $\mathbb{S}$: the coordinate-aligned $\mathbb{O}$, and Gresnigt's $\mathbb{O}_1, \mathbb{O}_2$. It establishes this as genuine *discrete* relativity — the mechanism is not an artifact of one coordinate choice — but leaves the *continuous* case explicitly open: whether $\mathbb{O}$ and $\mathbb{O}_1$ are related by the full, 14-dimensional continuous group G_2, of which only a finite discrete subgroup had been tested, exhaustively, at 4032 combinations, with zero matches. This note resolves that question.

2. A verified basis for octonion derivations

The derivations of the octonions, forming the Lie algebra $\mathfrak{g}_2$, are constructed from left- and right-multiplication operators $L_a(x) = ax, R_a(x) = xa$ (Schafer, *An Introduction to Nonassociative Algebras*):

$$D_{a,b} = [L_a, L_b] + [R_a, R_b] + [L_a, R_b].$$

Proposition 1. $D_{a,b}$ satisfies the Leibniz rule $D_{a,b}(xy) = D_{a,b}(x)y + xD_{a,b}(y)$ for all $a, b, x, y \in \mathbb{O}$.

Verified directly, for all 21 pairs D_{e_i,e_j}, $1 \le i < j \le 7$, across all 64 ordered basis pairs (x, y) each, without exception.

Proposition 2. The 21 matrices D_{e_i,e_j} have rank exactly 14.

Verified directly via singular value decomposition: the 21 singular values split cleanly into 14 equal, nonzero values and 7 exactly zero, with no ambiguity in the cutoff. The 14 right singular vectors corresponding to the nonzero values were reshaped into an orthonormal basis of 8×8 matrices and independently re-verified against the Leibniz rule; all 14 pass.

3. A structural trap, diagnosed

An initial attempt to test whether $\mathbb{O}$ and $\mathbb{O}_1$ are related by some continuous element of G_2 — diagonally extending each of the 14 generators to 16×16 block-diagonal matrices acting identically on both Cayley–Dickson halves, exponentiating linear combinations, and searching numerically for a match — produced a suspicious result: the distance between the transformed $\mathbb{O}$ and the fixed target $\mathbb{O}_1$ was found to be *exactly* constant, $2\sqrt{2}$, for every value of the 14 parameters tested, across thousands of random samples.

This is not a search failure but a structural necessity, and it is worth stating precisely why. $G_2 = \mathrm{Aut}(\mathbb{O})$ consists, by definition, of linear bijections of $\mathbb{O}$ onto itself. Any element of G_2 applied to $\mathbb{O}$'s own basis therefore maps the *span* of $\mathbb{O}$ onto itself exactly — it can only rotate which vectors span that fixed 8-dimensional space, never move the span elsewhere. Testing whether G_2, constructed from $\mathbb{O}$'s own structure constants, can move $\mathbb{O}$'s span to a different subspace was, by construction, guaranteed to fail regardless of the target — a fact that could have been anticipated from the definition of G_2 alone, but was only recognized after the computation exposed it.

This does not by itself resolve the original question. It shows only that *this specific diagonal construction* of G_2 cannot connect $\mathbb{O}$ and $\mathbb{O}_1$ — leaving open whether it captures all of G_2's action on $\mathbb{S}$, or only a restricted part of it.

4. Closing the gap: the diagonal extensions are complete

Two further facts, both independently verified, resolve this.

Proposition 3. Each of the 14 diagonal extensions $D \mapsto \mathrm{diag}(D, D)$ is a genuine derivation of the full 16-dimensional sedenion algebra $\mathbb{S}$ — not merely of $\mathbb{O}$.

Verified directly: for all 14 generators, the Leibniz rule was checked against $\mathbb{S}$'s complete, non-associative multiplication table across all 256 ordered basis pairs; all pass, without exception. The 14 extended 16×16 matrices were also confirmed linearly independent (rank 14).

Proposition 4. $\mathrm{der}(\mathbb{S})$, the full Lie algebra of derivations of $\mathbb{S}$, is exactly 14-dimensional.

This was computed from first principles, independent of any citation of $\mathrm{Aut}(\mathbb{S}) = G_2 \times S_3$ (Wilmot): the Leibniz condition $M(xy) = M(x)y + xM(y)$ for a general 16×16 matrix M was expressed as a linear system in M's 256 unknown entries — one equation per output coordinate, for each of $\mathbb{S}$'s 256 ordered basis pairs, yielding 4096 constraints. The rank of this constraint matrix was computed directly: 242, leaving a null space — the space of all derivations of $\mathbb{S}$ — of dimension exactly $256 - 242 = 14$.

Corollary. The 14 diagonal extensions of §2, being linearly independent elements of a space proven to be exactly 14-dimensional, form a complete basis for $\mathrm{der}(\mathbb{S})$. No derivation of $\mathbb{S}$ exists beyond their linear combinations.

5. The complete resolution

Since the 14 tested generators exhaust $\mathrm{der}(\mathbb{S})$ entirely (§4), the constancy observed in §3 is not a limitation of the search — it is a complete statement about the entire connected component of G_2's action on $\mathbb{S}$. No continuous deformation from the identity, however parametrized, can move $\mathbb{O}$'s span toward $\mathbb{O}_1$'s.

Combined with the exhaustive discrete check of the full S_3 factor (identity and both powers of ψ, already reported in the companion paper) against the full 1344-element discrete subgroup of G_2 — and given $\mathrm{Aut}(\mathbb{S}) = G_2 \times S_3$ is a direct product, so the two factors' actions commute — every element of $\mathrm{Aut}(\mathbb{S})$ has now been accounted for.

Theorem. $\mathbb{O}$ is not related to $\mathbb{O}_1$ by any element of $\mathrm{Aut}(\mathbb{S})$.

The identical test was repeated for $\mathbb{O}_2$: the same constant distance, $2\sqrt{2}$, was found for every sampled parameter value across all three S_3 choices, extending the result: $\mathbb{O}$ is not related to $\mathbb{O}_2$ by any element of $\mathrm{Aut}(\mathbb{S})$ either.

Remark ($\mathbb{O}_2$ and $\mathbb{O}_3$ follow by composition, and were independently verified regardless). *Generation Is Not Color* establishes genuine automorphisms $\rho : \mathbb{O}_1 \to \mathbb{O}_2$ and $\rho' : \mathbb{O}_1 \to \mathbb{O}_3$ — drawn from the discrete part of the G_2 factor, not from S_3, which stabilizes each $\mathbb{O}_i$ individually rather than relating them. Given these, the result for $\mathbb{O}_1$ already forces the results for $\mathbb{O}_2$ and $\mathbb{O}_3$: were some τ to relate $\mathbb{O}$ to $\mathbb{O}_2$, then $\rho^{-1} \circ \tau$ would relate $\mathbb{O}$ to $\mathbb{O}_1$, contradicting the theorem; the identical argument via ρ' rules out a relation to $\mathbb{O}_3$. Both were nonetheless checked directly, exactly as $\mathbb{O}_2$ was checked independently above despite this logical redundancy: $\mathbb{O}_3 = \{0, 3, 4, 7, 8, 11, 12, 15\}$ was tested against $\mathbb{O}$ with the identical discrete (4032-combination) and continuous searches, finding zero matches and the same constant distance $2\sqrt{2}$ across all sampled parameters. $\mathbb{O}$ is not related to any of Gresnigt's three octonion subalgebras by any element of $\mathrm{Aut}(\mathbb{S})$.

Corollary (the complements are equally disconnected). Since $g \in G_2$ is, by definition, an automorphism of $\mathbb{O}$, it maps $\mathbb{O}$ onto itself — and by the identical argument, $\mathrm{diag}(g, g)$ maps $\mathbb{O} \cdot e_8$ (the copy of $\mathbb{O}$ embedded in the second Cayley–Dickson slot) onto itself as well. The same completeness result of §4 therefore applies without modification: this was verified directly, with the discrete 4032-combination search finding zero matches and the continuous distance found constant at $2\sqrt{2}$ across all sampled parameters, for $\mathbb{O} \cdot e_8$ against $\mathbb{O}_1$'s, $\mathbb{O}_2$'s, and $\mathbb{O}_3$'s complements alike. $\mathbb{O} \cdot e_8$ is not related to any of the three complements by any element of $\mathrm{Aut}(\mathbb{S})$.

6. Discussion

This does not weaken the discrete relativity result of the companion paper — non-closure, the Cayley identity, faithfulness, and irreducibility genuinely recur, identically, across $\mathbb{O}$,

$\mathbb{O}_1, \mathbb{O}_2$, and $\mathbb{O}_3$. (For $\mathbb{O}_3$, whose complement was not separately run through the companion paper's four-property classification battery, this follows by transport: ρ' is a verified automorphism of $\mathbb{S}$ carrying $\mathbb{O}_1$ to $\mathbb{O}_3$ (*Generation Is Not Color*), and automorphisms preserve all four properties exactly.) What is now resolved is the *character* of that recurrence, and it is fully symmetric: neither $\mathbb{O}$ nor $\mathbb{O} \cdot e_8$ is related to any of $\mathbb{O}_1, \mathbb{O}_2$, $\mathbb{O}_3$, or their complements by any element of $\mathrm{Aut}(\mathbb{S})$. This is not gauge equivalence: $\mathbb{O}$ and $\mathbb{O}_1$ are not one underlying structure viewed through different coordinates, related by a symmetry the way $\mathbb{O}_1, \mathbb{O}_2$, and $\mathbb{O}_3$ are (companion paper, *Generation Is Not Color*). They are provably distinct, independently existing subalgebras of $\mathbb{S}$ that happen, without any connecting symmetry on either side, to share the exact same algebraic signature — four complete, self-contained instances of the entire Operator Mismatch structure, not one instance viewed four ways.

Scope. This resolves the relationship of $\mathbb{O}$ to all three of $\mathbb{O}_1, \mathbb{O}_2$, and $\mathbb{O}_3$. Whether $\mathbb{O}_1, \mathbb{O}_2$, and $\mathbb{O}_3$ are related to one another by the continuous part of G_2 (as opposed to the discrete color automorphism already established in *Generation Is Not Color*) was not tested here.

References

- Companion papers: *The Sedenion Split $\mathbb{S} = \mathbb{O} \oplus \mathbb{O} \cdot e_8$: Necessity, Classification, and Discrete Relativity*; *Generation Is Not Color*.
- Schafer, R. D., *An Introduction to Nonassociative Algebras*, Academic Press (1966); Dover reprint.
- Wilmot, G. P., "Automorphisms of Sedenions," arXiv:2512.07210.

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